

CE 513: Advanced Foundation Engineering

Course Code	CE 513
Course Name	Advanced Foundation Engineering
L-T-P-C	3-0-0-3
Prerequisites	CE402 or Equivalent
Intended for	UG and PG
Distribution	Discipline Elective
Approval	36th BoA (Sep 2020); 28th Senate (Nov 2020), in the B.Tech Civil Engineering curriculum

Course Contents

- **Bearing Capacity Theories for Shallow Foundation**
Foundation classification, choice of foundations; Field investigations, soil data and design parameters; Bearing capacity theories, influence of water table, soil compressibility, inclined loading, one-way and two-way eccentricity on the bearing capacity of soil. [4 Hours]
- **Bearing Capacity for Special Cases**
Bearing capacity for special cases of shallow foundations: foundations resting on layered soil, foundations supported by a soil with rigid base, foundations on sloping ground, closely placed foundations. [4 Lectures]
- **Allowable Bearing Capacity and Settlement of Foundations**
Vertical stress increase in a soil mass caused by different types of foundation load; Elastic settlement of foundations, settlement of sandy soil: use of strain influence factor and field test data; Consolidation settlement; Allowable bearing capacity from field load test; Rigid and flexible foundations, contact pressure and settlement distribution beneath footings; Construction issues such as dewatering methods and underpinning of shallow foundations; Foundations on difficult soil: collapsible soils, expansive soils, and sanitary landfills. [5 Lectures]
- **Mat Foundations**
Combined footings, mat/raft foundation; Bearing capacity and settlement of mat foundation; Design of mat foundation: rigid and flexible methods; Buoyancy raft or floating foundation; Skirted foundation. [7 Lectures]
- **Pile Foundations**
Deep foundations; Load transfer mechanism for pile foundation, estimation of pile capacity, settlement of pile; Small strain and large strain dynamic tests in piles; Pile foundation design subjected to compressive load, uplift, lateral load; Design of pile groups; Pile raft; Concept of negative skin friction; Construction issues related to pile foundation, effect of adjacent constructions on existing pile foundation. [18 Lectures]
- **Other Types of Deep Foundations**
Well foundations, classification, physical characteristics: shape, size, scour depth, components of well foundations, allowable bearing pressure, forces acting on well foundation and its lateral stability; An introduction to caisson foundation: mechanism, type and construction. [4 Lectures]

Text Books

1. B.M. Das, **Principles of Foundation Engineering**, Cengage Learning, 2013.
2. N.N. Som and S.C. Das, **Theory and Practice of Foundation Design**, Prentice-Hall of India Pvt.Ltd, 2004.

References

1. Joseph Bowles, **Foundation Analysis and Design**, McGraw Hill Education, 2017.
2. J.H. Atkinson, **The Mechanics of Soils and Foundations**, CRC Press, 2007.
3. Muni Budhu, **Soil Mechanics and Foundations**, John Wiley & Sons, Inc., 2010.
4. B.M. Das, **Shallow Foundations: Bearing Capacity and Settlement**, CRC Press, 2019.
5. Michael Tomlinson and John Woodward, **Pile Design and Construction Practice**, CRC Press, 2014.